

TMAIT — Traffic Management Plan

Generated by A.T.O.M | Grounded in BC TMM 2020 (Traffic Management Manual for Work on Roadways)

Job: Watermain repair - Lonsdale Ave **Status:** APPROVED **Model:** gpt-5.2

Location: Lonsdale Ave & 13th St, North Vancouver

1. Location Summary

Urban arterial intersection at Lonsdale Ave & 13th St, North Vancouver with high traffic volumes, bus route operations, and marked pedestrian crossings; 4-lane multilane undivided roadway context with frequent turning and curbside activity.

2. Event / Works Type

Utility work — long-duration daytime lane closure (multilane undivided roadway, urban arterial)

3. Duration

3 days (2026-08-01 to 2026-08-03), daily 07:00–17:00; all temporary signs/devices to be covered or removed outside work hours unless an ongoing hazard remains.

4. Road & Lane Closures

Close 1 right-hand travel lane on Lonsdale Ave through the work area (approx. 60 m work area length) with an upstream merge taper and downstream taper; maintain remaining lanes open to traffic. Maintain intersection operations; ensure lane closure does not block bus stop access where feasible and does not obstruct crosswalks/curb ramps. Install barricades at each end of the work activity area for long-duration work.

5. Detours

No detour required — remaining lanes on this 4-lane urban arterial can accommodate traffic with a right-lane closure using a standard merge taper and FAB in caution mode; access for local traffic, buses, and pedestrians maintained with signed/segregated work area.

6. Signage Schedule

Sign	Location	Spacing	Notes
C-130-R RIGHT LANE CLOSED AHEAD	Upstream advance warning on Lonsdale Ave in the affected direction, first sign encountered; place on right shoulder/curb line clear of bus stop operations and not obscured by parked vehicles	40	Use high-intensity retroreflective sheeting; mount to meet minimum height/offset; ensure not obscured (urban parking/bus activity).
C-018-1A CONSTRUCTION AHEAD	Upstream of taper, second sign encountered in advance warning area	40	Long-duration work sign; ensure visibility and not obscured by roadside activity.

C-053 LANE CLOSURE ARROW	At/near start of taper or on shadow vehicle at taper start (approach side), directing merge left	N/A	Permitted substitution for FAB at ≤ 60 km/h only if meeting Section 8 option criteria; for this high-volume arterial, provide a flashing arrow board (FAB) in caution mode as the primary lane-closure device.
C-086-1 THANK YOU RESUME SPEED	Downstream of work area after downstream taper (end of lane closure influence)	N/A	Place where drivers have fully cleared devices and normal operations resume.

7. Step-by-Step Setup

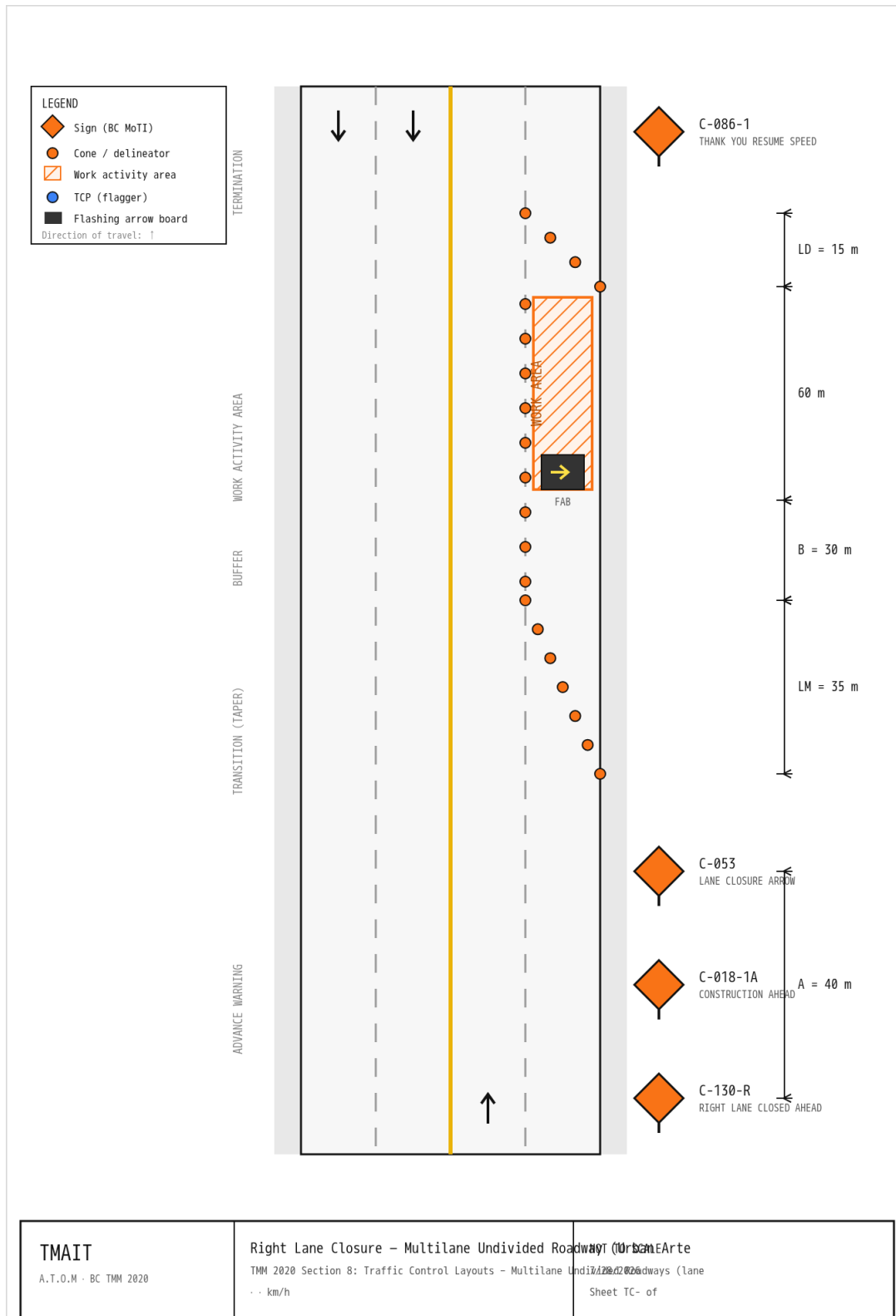
- 1 Confirm limits: identify exact lane-closure extents on Lonsdale Ave, maintain clear intersection sightlines, confirm bus stop locations and pedestrian crossings/curb ramps near 13th St.
- 2 Pre-installation checks: verify posted speed 50 km/h, confirm lane widths and available curb/shoulder space for signs without obstructing sidewalks or crosswalks; ensure all signs/devices meet retroreflectivity and are in good condition.
- 3 Deploy upstream advance warning signs in order (farthest first): place C-130-R then C-018-1A at Table B Distance A spacing for 50 km/h (40 m) approaching the work zone in the affected direction.
- 4 Position shadow/work vehicle on approach side with 360° flashing light and 4-way flashers operating; activate flashing arrow board (FAB) in caution mode for the lane closure.
- 5 Install merge taper: deploy cones/tubular markers for a 35 m merge taper (LM) for 50 km/h with device spacing 10 m; ensure taper aligns traffic smoothly out of the right lane.
- 6 Install buffer space: provide 30 m buffer (B) between end of taper and start of work activity area; keep buffer free of workers/materials.
- 7 Delineate work area: place cones/tubular markers at 10 m spacing along the closed right lane for the 60 m work area length; add rigid barricades at each end of the work activity area (long-duration requirement).
- 8 Install downstream taper: provide 15 m downstream taper (LD) with cones at 10 m spacing to transition back to normal lane configuration.
- 9 Place C-086-1 downstream beyond the downstream taper where traffic has re-established normal flow.
- 10 Pedestrian/bus accommodations: maintain accessible pedestrian routing around any sidewalk impacts (do not block curb ramps/crosswalks); if a bus stop is within the closure, coordinate temporary bus stop relocation or maintain bus pull-in/out with safe delineation as directed by the Road Authority/Transit operator.
- 11 Operational checks: drive-through/observe both approaches for sign conspicuity, queue development, and conflicts with turning movements at 13th St; adjust sign locations slightly as needed while maintaining Table B A spacing intent.
- 12 Daily close-out: at end of each shift, remove or cover temporary signs/devices when work is not underway and no continuing hazard exists; ensure excavations/fixtures are plated and the closed lane is either fully reopened or securely barricaded per Road Authority direction.

8. Safety Considerations

High AADT urban arterial with frequent stops/turning requires positive guidance: use FAB in caution mode for lane closure and maintain clear, consistent delineation through taper, buffer, and work area. Maintain a

30 m buffer space (B) and keep it clear of workers/equipment. Provide barricades at each end of the work activity area for long-duration work. Manage bus route hazard by ensuring signs are not placed where buses stop/turn and by maintaining predictable lane geometry; coordinate any bus stop impacts. Manage pedestrian crossing hazard by keeping crosswalks and curb ramps unobstructed, maintaining accessible pedestrian paths, and placing devices to prevent pedestrians from entering the work area. Ensure all work/shadow vehicles display 360° flashing light and 4-way flashers. TCPs not planned for a single-lane closure on a 4-lane undivided arterial; if queues become excessive or if turning movements/pedestrians require intermittent stoppages, add a TCP with appropriate advance signing (e.g., C-001-1 and optional C-029) and ensure any TCP station illumination if operating in low light.

Traffic Control Layout — TMM 2020 Schematic (Sheet TC-1 of 1)



NOTE: Provide FAB in caution mode at taper approach; install 35 m merge taper (10 m cone spacing), 30 m buffer, 60 m work area with barricades at each end, and 15 m downstream taper.

9. TMM 2020 Compliance Citations

TMM 2020 Section 8 (Multilane Undivided Roadways), 8.2 Typical Construction Speed Zone Signing — p.91 — For long-duration work, use C-018-1A CONSTRUCTION AHEAD; a flashing arrow board (FAB) shall be used when a lane is closed; all work/shadow vehicles shall have 360-degree flashing light and 4-way flashers. Plan uses C-018-1A, FAB in caution mode, and specifies vehicle lighting/flashers.

TMM 2020 Section 9 (Multilane Divided Roadways) general standards excerpt — p.123 — A flashing arrow board (FAB) shall be used when a lane is closed; barricade required in last affected lane on approach side for long-duration work; long-duration uses C-018-1A. Plan includes FAB and barricades at ends/approach side and uses C-018-1A.

TMM 2020 Section 8 options — Figure 8.10 notes — p.108 — If speed limit is ≤ 60 km/h, upstream C-130-R with distance tab may be omitted under certain conditions; substitution guidance for FAB at ≤ 60 km/h using C-053 plus lights. Plan retains C-130-R for high-volume arterial and uses FAB as primary, with C-053 as optional supplemental at taper.

TMM 2020 Section 7 TCP guidance/standards excerpt — p.64 — When TCPs are directing traffic, construction speed limit shall be ≤ 70 km/h; distance between TCP and C-001-1 should not exceed 150 m; C-029 Prepare to Stop is an option. Plan does not require TCPs for this closure but specifies conditions to add TCPs with compliant signing and placement if needed.

TMM 2020 Section 7.6 Work in Parking Lane – Urban Area — p.12 — Advance warning signs shall not be obscured by vehicles or other devices. Plan places warning signs to avoid obstruction from urban curbside activity (parking/bus operations) and requires visibility checks.

10. Site Diagram Reference (coordinates)

Map centre: 49.2827, -123.1207 — interactive diagram available in TMAIT workspace.

Type	Label	Lat	Lng
sign	C-130-R RIGHT LANE CLOSED AHEAD (NB)	49.28198	-123.12070
sign	C-018-1A CONSTRUCTION AHEAD (NB)	49.28234	-123.12070
sign	C-053 LANE CLOSURE ARROW (at taper)	49.28252	-123.12070
cone	Taper cones start	49.28258	-123.12070
cone	Taper cones mid	49.28268	-123.12070
cone	Taper cones end	49.28276	-123.12070
barrier	Barricade (upstream end of work area)	49.28283	-123.12070
work_zone	Work zone (watermain repair)	49.28290	-123.12070
barrier	Barricade (downstream end of work area)	49.28308	-123.12070
cone	Downstream taper start	49.28314	-123.12070
cone	Downstream taper end	49.28322	-123.12070
sign	C-086-1 THANK YOU RESUME SPEED	49.28334	-123.12070

Reviewer Feedback

TEST_ pytest approval — signage layout correct